



3D-Printed Bones

A ceramic-based ink could allow surgeons to 3D-print bone parts, complete with living cells.
<http://digit.in/feb21-05>



Cellular Autofluorescence

Scientists observe live cells sensing and responding to magnetic fields for the first time.
<http://digit.in/feb21-06>

New Biometric Indicators

How will tech measure you up next?

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iometric indicators are used to uniquely identify people, and allow access to buildings, facilities and

systems. Biometric authentication is widespread, and is even used in smartphones, ATM kiosks, banks, airports, hospitals, and office complexes. The biometric modes in use include fingerprints, faces, voice signals, eyes, ears, DNA profiles, EEG and ECG signals. For a biometric indicator to be useful, it has to have certain characteristics, namely universality, uniqueness, permanence, collectability, performance, acceptability, and circumvention. Here permanence means a lack of variation across life, so that the indicator should not change between

a baby and an old man. A fingerprint will not, but the individual's voice most certainly will. Performance is how good the biometric indicator is in real world use. In theory, iris recognition is great, but in practice, because of the low quality of imaging systems, it is very unreliable and often locks people out. Collectability is the ease at which the particular indicator can be measured. While it may be easy to recognise facial characteristics through any available video camera, it is considerably more painstaking to use DNA-based biometric authenticators. Out of all these DNA has been considered to be the most secure, but it is possible in theory to gather and amplify DNA, or even engineer new DNA and then use it to bypass the authentication so none of the modes are considered foolproof. To make existing systems more secure, one of the approaches is to use multimodal authentication techniques. This means that a fingerprint is checked in tandem with the iris. Any new biometric indicator would have to be

easy to collect, integrate into multimodal systems, and preferably be able to run off your smartphone with all the data stored locally.

PATTERNS IN THE VEINS

Knowing something as good as the back of your hand is actually a misleading statement, as the back of the hand is one of the most complicated parts in the human body. There are all sorts of crisscrossing tendons, veins, muscles, and fourteen bones in each hand. As the human body grows, the venation pattern in each individual's hands are unique, and this can be potentially captured and read by existing multimodal systems that for example check for fingerprints, palm prints and hand geometry all through a single "hand scan". Researchers from the University at Buffalo have used photoacoustic tomography to make a three dimensional map of the veins in the hands of an individual, and then use it to match against a saved sample. The approach uses a combination of both light and sound.